Hybrid-Attention and Depth-Mixing Kernels

Deep Dive into Kimi K3 — Chapter 35

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Foundation Books Project

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Goal: type every retained object by owner and last reader, take Gated MLA from logical record to tiles and dispatch, build Full and two-phase Block AttnRes kernels, shard, overlap, and differentiate them — and keep every claim in scope.

Roadmap

Typed state and the hybrid sequence path

Depth kernels: Full, Block, and two phases

Sharding, overlap, backward, and evidence

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The model schedule selects the operator; the support envelope selects only its implementation.

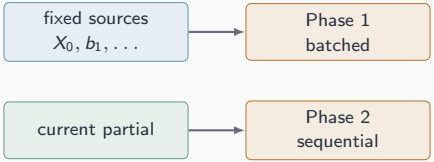
Depth kernels: Full, Block, and two phases

Carry the whole summary across the phase boundary



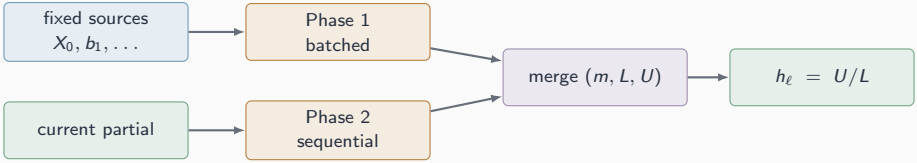
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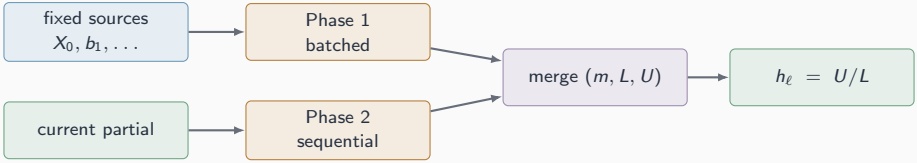
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- Toy fixture: Full 66 versus Block 29 source interactions — a change of representation, not a same-model speedup. [DERIVED: one token row]

Sharding, overlap, backward, and evidence

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Diagnose in dependency order. Next: Chapter 36's MoE Kernels and Routed Computation.